FIT1058 — Chapter 5: Predicate Logic

Exercise Solutions

Alston

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Exercise 5

(six degrees of separation)

Consider the binary predicate knows from p. 68 in § 2.14. The domain of each of its arguments is the set of all people.

It has been claimed that, in the human social network, the distance between any two people is at most 6. (See p. 70 in § 2.15.)

Write this claim in predicate logic, using just the predicate knows.

Exercise 7

For each of the following predicate logic statements, (i) identify the predicates, functions, variables (including a suggested domain for each) and constants used; (ii) state whether or not it is a proposition, and if it is, whether it is true or false.

(a) $\forall x \; x^2 \geq 0$

(b) $x^2 < 0$

(c) $\forall x \; 2x = x^2$

(d) $x < 0 \Rightarrow \forall y \big((0 < y) \Rightarrow (x < y)\big)$

(e) $\forall x \, \forall z \, \exists y \; (x < y) \wedge (y < z)$

(f) $\exists x \, \exists y \; \neg(x \Rightarrow y)$

Exercise 8

Consider the following predicate logic statement.

$$\forall A \, \forall B \; (A \subseteq B) \Rightarrow (\mathcal{P}(A) \subseteq \mathcal{P}(B))$$

What predicates, functions and variables does this statement use? Restate, in words, the theorem that is being stated here.

Exercise 9

Suppose that

- W is a variable whose domain is the set of all English words,
- L is a variable whose domain is the set of all English letters,

- WordContainsLetter is a binary predicate which takes an English word as its first argument, an English letter as its second argument, and is True whenever that word contains that letter. For example,

$$\text{WordContainsLetter}(\text{quizzical}, z) \text{ is True,}$$

$$\text{WordContainsLetter}(\text{quizzical}, e) \text{ is False.}$$

(a) Write the statement of Theorem 15 in predicate logic.

(b) Now suppose you also have a unary predicate Vowel whose domain is the set of English letters and which is True when its argument is a vowel. Now write the statement of Theorem 15 in predicate logic again, but using this new predicate to write it more compactly this time.

Solution. Write WordContainsLetter(W, L) as $W(W, L)$.

(a)

$$\forall W\, \exists L\, \Big(\big(L = a \vee L = e \vee L = i \vee L = o \vee L = u\big) \wedge W(W, L) \Big) \;\vee\; W(W, y)$$

(b) Writing Vowel(L) as $V(L)$:

$$\forall W\, \exists L\, \Big(V(L) \wedge W(W, L) \Big) \;\vee\; W(W, y)$$

or, with the disjunction brought inside the scope of $\exists L$:

$$\forall W\, \Big(\exists L\, \big(V(L) \wedge W(W, L) \big) \;\vee\; W(W, y) \Big)$$

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Exercise 10

Suppose you have the positive integer properties Even and Prime, which are true precisely for even numbers and prime numbers respectively, and that you also have the binary predicate $\leq$ on $\mathbb{N}$. Write a predicate logic expression for the statement

There are infinitely many odd prime numbers but only finitely many even prime numbers.

Solution. Write Even(n) as $E(n)$ and Prime(n) as $P(n)$. By the domain assumption, any element that is not even is odd, so "odd" is $\neg E(n)$, and "odd prime" is $P(n) \wedge \neg E(n)$.

Infinitely many odd primes. There is no largest odd prime – for any bound b, some odd prime exceeds it:

$$R = \forall b\, \exists n\, \big(P(n) \wedge \neg E(n) \wedge b \leq n \big)$$

Only finitely many even primes. There is a largest even prime m, and every even prime is at most m:

$$S = \exists m\, \Big(P(m) \wedge E(m) \;\wedge\; \forall n \big(P(n) \wedge E(n) \Rightarrow n \leq m \big) \Big)$$

Final statement.

$$R \wedge S$$

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Exercise 11

The ternary predicate Date is defined for any $(d, m, y) \in \mathbb{N} \times \mathbb{N} \times \mathbb{N}$, and is True if (d, m, y) represents a valid date in day-month-year format in the Gregorian calendar, and is False otherwise.

Using Date and $<$, write an expression in predicate logic which is True if and only if (d_1, m_1, y_1) and (d_2, m_2, y_2) are both valid dates and the first date comes chronologically before the second.

Solution. Write Date(d, m, y) as $D(d, m, y)$. (d_1, m_1, y_1) comes chronologically before (d_2, m_2, y_2) exactly when

$$y_1 < y_2, \quad \text{or} \quad y_1 = y_2 \wedge m_1 < m_2, \quad \text{or} \quad y_1 = y_2 \wedge m_1 = m_2 \wedge d_1 < d_2.$$

So the answer is

$$D(d_1, m_1, y_1) \wedge D(d_2, m_2, y_2) \wedge \Big((y_1 < y_2) \vee \big((y_1 = y_2) \wedge (m_1 < m_2) \big) \vee \big((y_1 = y_2) \wedge (m_1 = m_2) \wedge (d_1 < d_2) \big) \Big)$$

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Exercise 12

Suppose you have

- the positive integer predicate $\leq$;
- the function $t \colon \{\text{programs}\} \times A^* \to \mathbb{N}$ defined for all programs P and input strings $x \in A^*$ (where A is an alphabet, § 1.5) by

$$t(P, x) = \text{the time taken by program } P \text{ when run on input } x;$$

- the function len: $A^* \to \mathbb{N}_0$ defined for any string $x \in A^*$ by

$$\text{len}(x) = \text{the length of the string } x;$$

- binary functions for multiplication and exponentiation on the positive integers;
- a string variable $x \in A^*$;
- positive integer variables c and k.

Write the following statement in predicate logic.

> There exist integers c and k such that, for every input string x, the time taken by program P on input x is at most the product of c and the k-th power of the length of x.

Exercise 13

There are many theorems that assert that there is a unique object that satisfies certain conditions. In this exercise, we look at how to make such statements using predicate logic.

Suppose you want to say that there is a unique x with property $P(x)$. It's not enough to just write

$$\exists x \; P(x),$$

because that only enforces existence without guaranteeing uniqueness. Sometimes, people use "$\exists!$" as a shorthand for "there exists a unique". With that shorthand,

$$\exists! x \; P(x)$$

means

There exists a unique x such that $P(x)$ holds.

How would you write this statement just using the tools available to us in predicate logic, i.e., without using "!"?

Solution.

$$\exists! x \ P(x) \ \equiv \ \exists x \ \Big(P(x) \wedge \big(\forall y \ (P(y) \Rightarrow x = y)\big)\Big)$$

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Exercise 14

(a) Prove that

$$\exists X\big(P(X) \wedge Q(X)\big) \text{ implies } \big(\exists X \ P(X)\big) \wedge \big(\exists X \ Q(X)\big)$$

(b) Prove that

$$\forall X\big(P(X) \vee Q(X)\big) \text{ is implied by } \big(\forall X \ P(X)\big) \vee \big(\forall X \ Q(X)\big)$$

in two ways: (i) reason it through directly, (ii) use the result from part (a) and what you've learned about the relationship between existential and universal quantifiers.